

Year 7 · Chapter 12 · Ratios, rates and time: answers (teacher copy)

12.01 Ratios (booklet p. 3)

WE DO examples (teacher)

- Column 1, E1: 10 green, 6 red: green to red $\rightarrow 10 : 6$
- Column 1, E2: 13 yellow, 9 blue: yellow to blue $\rightarrow 13 : 9$
- Column 2, E1: 16 girls to 14 boys $\rightarrow 16 : 14; \div 2: 8 : 7$
- Column 2, E2: 24 girls to 42 boys $\rightarrow 24 : 42; \div 6: 4 : 7$
- Column 3, E1: 120 g to 3 kg $\rightarrow 3 \text{ kg} = 3000 \text{ g}; 120 : 3000; = 1 : 25$
- Column 3, E2: 20c to \$3 $\rightarrow \$3 = 300 \text{ c}; 20 : 300; = 1 : 15$

Questions (in order: Easy, Medium, Challenging)

- | | | |
|------------|-------------|-------------|
| 1. 15 : 1 | 13. 5 : 13 | 25. 3 : 1 |
| 2. 7 : 7 | 14. 13 : 27 | 26. 1 : 9 |
| 3. 9 : 9 | 15. 3 : 9 | 27. 4 : 1 |
| 4. 2 : 17 | 16. 2 : 11 | 28. 4 : 5 |
| 5. 4 : 14 | 17. 2 : 2 | 29. 5 : 24 |
| 6. 1 : 4 | 18. 8 : 14 | 30. 9 : 400 |
| 7. 10 : 12 | 19. 3 : 4 | 31. 9 : 200 |
| 8. 10 : 5 | 20. 4 : 7 | 32. 1 : 40 |
| 9. 6 : 16 | 21. 5 : 8 | 33. 1 : 15 |
| 10. 5 : 5 | 22. 7 : 3 | 34. 3 : 40 |
| 11. 7 : 9 | 23. 2 : 3 | 35. 1 : 25 |
| 12. 4 : 4 | 24. 1 : 5 | |

12.02 Equivalent and simplified ratios

(booklet p. 4)

WE DO examples (teacher)

- Column 1, E1: $36 : 27 \rightarrow 4 : 3$
- Column 1, E2: $4 : 8 \rightarrow 1 : 2$
- Column 2, E1: $3 : 4 = 9 : \square \rightarrow 3 \times 3 = 9; \square = 4 \times 3 = 12$
- Column 2, E2: $7 : 3 = \square : 18 \rightarrow 3 \times 6 = 18; \square = 7 \times 6 = 42$
- Column 3, E1: $0.5 : 0.2 \rightarrow \times 10: 5 : 2; = 5 : 2$
- Column 3, E2: $4 : 32 : 4 \rightarrow \text{HCF} = 4; = 1 : 8 : 1$

Questions (in order: Easy, Medium, Challenging)

- | | | |
|-------------|-------------|---------------|
| 1. 7 : 8 | 22. 11 : 10 | 43. 24 |
| 2. 4 : 7 | 23. 11 : 9 | 44. 14 |
| 3. 2 : 7 | 24. 7 : 2 | 45. 63 |
| 4. 2 : 1 | 25. 7 : 10 | 46. 18 |
| 5. 2 : 9 | 26. 5 : 4 | 47. 3 |
| 6. 4 : 7 | 27. 3 : 8 | 48. 24 |
| 7. 1 : 5 | 28. 9 : 8 | 49. 8 |
| 8. 8 : 7 | 29. 3 : 11 | 50. 6 |
| 9. 5 : 12 | 30. 64 | 51. 21 |
| 10. 4 : 3 | 31. 72 | 52. 7 : 2 : 7 |
| 11. 6 : 7 | 32. 24 | 53. 3 : 1 |
| 12. 10 : 11 | 33. 4 | 54. 1 : 6 |
| 13. 5 : 1 | 34. 12 | 55. 9 : 5 : 5 |
| 14. 7 : 4 | 35. 8 | 56. 9 : 5 : 9 |
| 15. 10 : 3 | 36. 4 | 57. 7 : 3 : 4 |
| 16. 7 : 2 | 37. 32 | 58. 1 : 4 |
| 17. 3 : 4 | 38. 42 | 59. 2 : 7 : 4 |
| 18. 12 : 7 | 39. 24 | 60. 9 : 7 |
| 19. 9 : 4 | 40. 16 | 61. 2 : 1 |
| 20. 4 : 5 | 41. 24 | |
| 21. 8 : 9 | 42. 25 | |

12.03 Ratio problems (booklet p. 5)

WE DO examples (teacher)

- Column 1, E1: \$60 in 4 : 1 $\rightarrow \$48, \12
- Column 1, E2: \$42 in 3 : 3 $\rightarrow \$21, \21

- Column 2, E1: \$34 in the ratio 1 : 1 $\rightarrow$ parts: $1 + 1 = 2$; one part: $34 \div 2 = 17$; \$17 and \$17
- Column 2, E2: \$50 in the ratio 5 : 5 $\rightarrow$ parts: $5 + 5 = 10$; one part: $50 \div 10 = 5$; \$25 and \$25
- Column 3, E1: Cordial : water = 4 : 8. How much water for 280 mL of cordial? $\rightarrow 4 \text{ parts} = 280 \text{ mL}; 1 \text{ part} = 70 \text{ mL}; \text{water: } 8 \times 70 = 560 \text{ mL}$
- Column 3, E2: Boys : girls = 3 : 8. There are 64 girls. How many students altogether? $\rightarrow 8 \text{ parts} = 64, 1 \text{ part} = 8$; boys: $3 \times 8 = 24$; total: $24 + 64 = 88$

Questions (in order: Easy, Medium, Challenging)

- | | | |
|----------------|----------------|-------------------|
| 1. \$18, \$27 | 15. \$20, \$20 | 29. \$30, \$30 |
| 2. \$10, \$6 | 16. \$6, \$8 | 30. \$68 and \$68 |
| 3. \$40, \$30 | 17. \$95, \$57 | 31. \$10 and \$50 |
| 4. \$68, \$68 | 18. \$21, \$7 | 32. \$7 and \$28 |
| 5. \$9, \$12 | 19. \$56, \$14 | 33. \$75 and \$75 |
| 6. \$27, \$9 | 20. \$14, \$14 | 34. \$14 and \$28 |
| 7. \$60, \$20 | 21. \$28, \$56 | 35. \$55 and \$33 |
| 8. \$54, \$18 | 22. \$64, \$48 | 36. \$28 and \$35 |
| 9. \$10, \$20 | 23. \$33, \$22 | 37. 210 mL |
| 10. \$13, \$39 | 24. \$8, \$32 | 38. 20 |
| 11. \$18, \$18 | 25. \$21, \$35 | 39. 240 mL |
| 12. \$55, \$55 | 26. \$36, \$27 | 40. 110 |
| 13. \$9, \$9 | 27. \$18, \$18 | 41. 360 mL |
| 14. \$26, \$39 | 28. \$36, \$24 | |

12.04 Rates (booklet p. 6)

WE DO examples (teacher)

- Column 1, E1: 273 m in 7 s $\rightarrow 39 \text{ m/s}$
- Column 1, E2: \$188 in 4 kg $\rightarrow \$47/\text{kg}$
- Column 2, E1: A car travels 525 km in 5 hours. Find its speed. $\rightarrow 525 \text{ km} \div 5 \text{ h}; = 105 \text{ km/h}$
- Column 2, E2: A car travels 872 km in 8 hours. Find its speed. $\rightarrow 872 \text{ km} \div 8 \text{ h}; = 109 \text{ km/h}$
- Column 3, E1: Car A: 35 km/h. Car B: 142 km in 3 h. Which is faster? $\rightarrow \text{B: } 142 \div 3 = 47.3 \text{ km/h}; \text{A: } 35 \text{ km/h}; \text{faster: B}$
- Column 3, E2: Car A: 36 km/h. Car B: 355 km in 4 h. Which is faster? $\rightarrow \text{B: } 355 \div 4 = 88.8 \text{ km/h}; \text{A: } 36 \text{ km/h}; \text{faster: B}$

Questions (in order: Easy, Medium, Challenging)

- | | | |
|-----------------|------------------|--------------|
| 1. 29 words/min | 16. 15 m/s | 31. 75 km/h |
| 2. \$52/kg | 17. \$41/kg | 32. 101 km/h |
| 3. 58 m/s | 18. 28 words/min | 33. 106 km/h |
| 4. 50 words/min | 19. 54 L/min | 34. 96 km/h |
| 5. 30 words/min | 20. 13 m/s | 35. 94 km/h |
| 6. 20 L/min | 21. 58 L/min | 36. 46 km/h |
| 7. 33 m/s | 22. 40 m/s | 37. 92 km/h |
| 8. 19 km/h | 23. 9 m/s | 38. 61 km/h |
| 9. 46 m/s | 24. 59 L/min | 39. A |
| 10. \$49/kg | 25. 48 km/h | 40. B |
| 11. \$41/kg | 26. \$6/kg | 41. B |
| 12. \$43/kg | 27. 12 words/min | 42. B |
| 13. 23 m/s | 28. 16 m/s | 43. B |
| 14. 19 m/s | 29. 53 km/h | 44. B |
| 15. 26 km/h | 30. 9 km/h | |

12.05 Best buys (booklet p. 7)

WE DO examples (teacher)

- Column 1, E1: 9 for \$74.16 $\rightarrow \$8.24$
- Column 1, E2: 6 for \$8.52 $\rightarrow \$1.42$
- Column 2, E1: 2 for \$6.16 or 4 for \$11.00 $\rightarrow \$6.16 \div 2 = \$3.08; \$11.00 \div 4 = \2.75 ; better: 4 for \$11.00

- Column 2, E2: 6 for \$6.60 or 5 for \$11.35 → $\$6.60 \div 6 = \1.10 ; $\$11.35 \div 5 = \2.27 ; better: 6 for \$6.60
- Column 3, E1: 300 g for \$4.97 or 1000 g for \$11.92 → 300 g: $\$4.97 \div 3 = \1.66 per 100 g; 1000 g: $\$11.92 \div 10 = \1.19 per 100 g; better buy: 1000 g
- Column 3, E2: 250 g for \$5.71 or 500 g for \$6.52 → 250 g: $\$5.71 \div 2.5 = \2.28 per 100 g; 500 g: $\$6.52 \div 5 = \1.30 per 100 g; better buy: 500 g

Questions (in order: Easy, Medium, Challenging)

- | | | |
|------------|------------------|-------------------|
| 1. \$4.79 | 17. \$2.99 | 33. 5 for \$15.75 |
| 2. \$7.27 | 18. \$6.34 | 34. 5 for \$15.05 |
| 3. \$6.98 | 19. \$4.19 | 35. 2 for \$4.90 |
| 4. \$5.87 | 20. \$2.26 | 36. 4 for \$7.12 |
| 5. \$5.36 | 21. \$6.07 | 37. 5 for \$16.55 |
| 6. \$5.49 | 22. \$7.60 | 38. 3 for \$6.60 |
| 7. \$5.80 | 23. \$7.72 | 39. 2 for \$5.98 |
| 8. \$5.17 | 24. \$5.15 | 40. 3 for \$7.05 |
| 9. \$3.33 | 25. \$5.75 | 41. 1000 g |
| 10. \$3.15 | 26. \$0.79 | 42. 1000 g |
| 11. \$3.32 | 27. \$5.16 | 43. 200 g |
| 12. \$2.16 | 28. \$0.84 | 44. 400 g |
| 13. \$5.32 | 29. \$2.39 | 45. 1000 g |
| 14. \$2.58 | 30. \$2.56 | 46. 1000 g |
| 15. \$8.07 | 31. 2 for \$3.14 | |
| 16. \$8.66 | 32. 3 for \$5.88 | |

12.06 Rate problems (booklet p. 8)

WE DO examples (teacher)

- Column 1, E1: 10 kg at \$5/kg → \$50
- Column 1, E2: 7 h at 10 km/h → 70 km
- Column 2, E1: 300 km at 60 km/h → $300 \div 60 = 5$ h
- Column 2, E2: 120 km at 40 km/h → $120 \div 40 = 3$ h
- Column 3, E1: Mia earns \$16 an hour. She works 7 hours a day for 4 days. How much does she earn? → one day: $16 \times 7 = \$112$; 4 days: $112 \times 4 = \$448$
- Column 3, E2: Mia earns \$27 an hour. She works 6 hours a day for 4 days. How much does she earn? → one day: $27 \times 6 = \$162$; 4 days: $162 \times 4 = \$648$

Questions (in order: Easy, Medium, Challenging)

- | | | |
|------------|-------------|-----------|
| 1. 320 km | 16. \$140 | 31. 4 h |
| 2. 135 km | 17. \$28 | 32. 4 h |
| 3. 385 km | 18. \$51 | 33. 6 h |
| 4. \$100 | 19. 240 km | 34. 2 h |
| 5. \$114 | 20. \$52 | 35. 6 h |
| 6. \$28 | 21. \$63 | 36. 1 h |
| 7. \$21 | 22. \$72 | 37. 3 h |
| 8. \$140 | 23. \$18 | 38. 5 h |
| 9. \$240 | 24. \$190 | 39. 6 h |
| 10. \$112 | 25. \$96 | 40. 1 h |
| 11. \$220 | 26. \$144 | 41. \$170 |
| 12. 720 km | 27. 345 km | 42. \$144 |
| 13. \$90 | 28. 900 km | 43. \$600 |
| 14. \$220 | 29. 1320 km | 44. \$240 |
| 15. \$24 | 30. 160 km | 45. \$544 |

12.07 Travel graphs (booklet p. 9)

WE DO examples (teacher)

- Column 1, E1: 120 km in 3 h → 40 km/h
- Column 1, E2: 120 km in 6 h → 20 km/h
- Column 2, E1: How far in the first hour? → $40 - 0 = 40$ km
- Column 2, E2: What was the total distance? → read the last point; = 120 km
- Column 3, E1: $s = 50$ km/h, $t = 4$ h. Find d. → $d = s \times t = 50 \times 4 = 200$ km

- Column 3, E2: $d = 80$ km, $t = 4$ h. Find s. → $s = d \div t = 80 \div 4 = 20$ km/h
- Column 3, E3: $d = 250$ km, $s = 50$ km/h. Find t. → $t = d \div s = 250 \div 50 = 5$ h

Questions (in order: Easy, Medium, Challenging)

- | | | |
|-------------|--------------|---------------------|
| 1. 45 km/h | 15. 20 km/h | 29. 45 km/h |
| 2. 20 km/h | 16. 10 km/h | 30. 90 km/h |
| 3. 40 km/h | 17. 10 km/h | 31. from 2 h to 3 h |
| 4. 20 km/h | 18. 80 km/h | 32. 20 km/h |
| 5. 80 km/h | 19. 40 km/h | 33. 40 km/h |
| 6. 45 km/h | 20. 90 km/h | 34. 24 km/h |
| 7. 60 km/h | 21. 60 km/h | 35. 240 km |
| 8. 45 km/h | 22. 10 km/h | 36. 60 km/h |
| 9. 100 km/h | 23. 15 km/h | 37. 4 h |
| 10. 15 km/h | 24. 40 km/h | 38. 120 km |
| 11. 10 km/h | 25. 100 km/h | 39. 50 km/h |
| 12. 90 km/h | 26. 80 km/h | 40. 2 h |
| 13. 80 km/h | 27. 20 km/h | |
| 14. 90 km/h | 28. 15 km/h | |

12.08 Time (booklet p. 10)

WE DO examples (teacher)

- Column 1, E1: 420 min = ____ h → 7 h
- Column 1, E2: 6 min = ____ s → 360 s
- Column 2, E1: 3 h 46 min + 2 h 15 min → 5 h 61 min; = 6 h 1 min
- Column 2, E2: 3 h 50 min + 2 h 20 min → 5 h 70 min; = 6 h 10 min
- Column 3, E1: 2 h 45 min = ____ h (decimal) → $45 \text{ min} \div 60 = 0.75$; = 2.75 h
- Column 3, E2: 5 h 24 min = ____ h (decimal) → $24 \text{ min} \div 60 = 0.4$; = 5.4 h

Questions (in order: Easy, Medium, Challenging)

- | | | |
|-------------|----------------|----------------|
| 1. 420 min | 15. 300 s | 29. 4 h 42 min |
| 2. 360 min | 16. 420 s | 30. 6 h 8 min |
| 3. 2 h | 17. 540 min | 31. 3 h 47 min |
| 4. 180 s | 18. 4 h | 32. 5 h 30 min |
| 5. 300 min | 19. 540 s | 33. 3 h 24 min |
| 6. 240 s | 20. 9 h | 34. 6.1 h |
| 7. 240 min | 21. 3 h | 35. 4.2 h |
| 8. 480 s | 22. 180 min | 36. 1.75 h |
| 9. 5 h | 23. 3 h 26 min | 37. 3 h 30 min |
| 10. 120 s | 24. 3 h 54 min | 38. 6 h 6 min |
| 11. 120 min | 25. 5 h 54 min | 39. 4.5 h |
| 12. 6 h | 26. 5 h 50 min | 40. 6.2 h |
| 13. 480 min | 27. 5 h 8 min | 41. 4.25 h |
| 14. 8 h | 28. 4 h 38 min | 42. 5.5 h |

12.09 12-hour and 24-hour time (booklet p. 11)

WE DO examples (teacher)

- Column 1, E1: 10:25 p.m. → 22:25
- Column 1, E2: 4:05 a.m. → 04:05
- Column 1, E3: 14:00 → 2:00 p.m.
- Column 2, E1: 11:40 p.m. → p.m.: add 12; = 23:40
- Column 2, E2: 12:55 p.m. → p.m.: 12 p.m. stays 12; = 12:55
- Column 3, E1: 80 min after 08:15 → $08:15 + 80 \text{ min}$; = 09:35; = 9:35 a.m.
- Column 3, E2: 50 min after 21:55 → $21:55 + 50 \text{ min}$; = 22:45; = 10:45 p.m.

Questions (in order: Easy, Medium, Challenging)

- | | | |
|----------|----------|-----------|
| 1. 12:00 | 5. 15:20 | 9. 00:10 |
| 2. 02:35 | 6. 02:00 | 10. 15:40 |
| 3. 11:10 | 7. 06:55 | 11. 16:55 |
| 4. 08:40 | 8. 16:00 | 12. 13:15 |

- | | | |
|----------------|-----------------------|------------------------|
| 13. 12:50 p.m. | 25. 2:05 p.m. | 37. 12:50 (12:50 p.m.) |
| 14. 3:45 a.m. | 26. 12:15 | 38. 04:55 (4:55 a.m.) |
| 15. 3:20 p.m. | 27. 23:15 | 39. 15:05 (3:05 p.m.) |
| 16. 5:45 p.m. | 28. 00:05 | 40. 12:45 (12:45 p.m.) |
| 17. 9:25 p.m. | 29. 00:30 | 41. 07:00 (7:00 a.m.) |
| 18. 12:10 p.m. | 30. 23:55 | 42. 17:05 (5:05 p.m.) |
| 19. 8:15 p.m. | 31. 12:35 | |
| 20. 3:45 p.m. | 32. 23:20 | |
| 21. 8:15 a.m. | 33. 12:50 | |
| 22. 5:00 a.m. | 34. 11:40 | |
| 23. 7:05 p.m. | 35. 00:40 | |
| 24. 2:10 p.m. | 36. 14:20 (2:20 p.m.) | |

12.10 Time differences (booklet p. 12)

WE DO examples (teacher)

- Column 1, E1: 1:10 p.m. to 2:40 p.m. → 1 h 30 min
- Column 1, E2: 9:20 a.m. to 10:30 a.m. → 1 h 10 min
- Column 2, E1: 16:25 to 21:30 → 16:25 → 17:00: 35 min; 17:00 → 21:30: 4 h 30 min; total: 5 h 5 min
- Column 2, E2: 15:30 to 18:45 → 15:30 → 16:00: 30 min; 16:00 → 18:45: 2 h 45 min; total: 3 h 15 min
- Column 3, E1: A flight leaves Perth at 2:35 p.m. and takes 4 h 50 min. What is the time in Sydney when it lands? → lands: 2:35 p.m. + 4 h 50 min = 7:25 p.m. Perth time; + 2 h for Sydney; = 9:25 p.m.
- Column 3, E2: A flight leaves Perth at 11:35 a.m. and takes 3 h 40 min. What is the time in Sydney when it lands? → lands: 11:35 a.m. + 3 h 40 min = 3:15 p.m. Perth time; + 2 h for Sydney; = 5:15 p.m.

Questions (in order: Easy, Medium, Challenging)

- | | | |
|---------------|---------------|-----------|
| 1. 3 h 5 min | 3. 3 h 35 min | 5. 2 h |
| 2. 1 h 25 min | 4. 5 h | 6. 35 min |

- | | | |
|----------------|----------------|----------------|
| 7. 1 h | 15. 1 h 5 min | 23. 3 h 50 min |
| 8. 1 h 50 min | 16. 45 min | 24. 4 h 25 min |
| 9. 2 h 35 min | 17. 1 h 15 min | 25. 3 h 45 min |
| 10. 4 h 40 min | 18. 2 h 45 min | 26. 8:55 p.m. |
| 11. 2 h 55 min | 19. 1 h 40 min | 27. 1:00 p.m. |
| 12. 1 h 5 min | 20. 1 h 35 min | 28. 5:05 p.m. |
| 13. 3 h | 21. 2 h 10 min | 29. 1:15 p.m. |
| 14. 30 min | 22. 1 h 20 min | |

12.11 Timetables (booklet p. 13)

WE DO examples (teacher)

- Column 1, E1: Bus 1 at the Library → 08:12
- Column 1, E2: Bus 2 at the Library → 08:42
- Column 2, E1: Bus 1: School to Library → 08:05 to 08:12; = 7 min
- Column 2, E2: Bus 2: Library to Shops → 08:42 to 09:03; = 21 min
- Column 3, E1: You must be at the Beach by 10:07. What is the latest bus you can catch from School? → Bus 3 reaches the Beach at 10:02; Bus 4 arrives 10:17: too late
- Column 3, E2: You reach the Library at 08:44. When is the next bus, and when does it reach the Pool? → next bus: Bus 3 at 09:27; reaches the Pool at 09:39

Questions (in order: Easy, Medium, Challenging)

- | | | |
|----------|------------|----------------|
| 1. 09:27 | 10. 10:03 | 19. 7 min |
| 2. 09:42 | 11. 08:47 | 20. 21 min |
| 3. 08:24 | 12. 09:17 | 21. 9 min |
| 4. 08:54 | 13. 10:02 | 22. 19 min |
| 5. 09:39 | 14. 10:17 | 23. 12 min |
| 6. 09:54 | 15. 9 min | 24. 23 min |
| 7. 08:33 | 16. 19 min | 25. 42 min |
| 8. 09:03 | 17. 12 min | 26. 10:03 a.m. |
| 9. 09:48 | 18. 23 min | 27. 30 min |